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Prepared for beginner Linux laptop users · September 2026

Backing Up Personal Data on a Linux Laptop

A beginner’s guide to image tools, snapshot tools, the 3-2-1-1-0 rule, and a ten-step workflow you can run tonight

What this is	A practical, evidence-backed walkthrough of how to protect personal data on a Linux laptop: which layer of backup solves which problem, the mistakes that silently destroy backups, and an exact ten-step setup using Restic.
Who it is for	People who have never run a real backup, or who have run a tool but never tried a restore. No prior storage experience assumed.
What it is not	Not a benchmark. No commands in this document were executed or timed; restore-time figures quoted from vendors are marketing copy, and distro package versions were not verified. See the Limitations section.

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1. Why you need this

The ArchWiki puts it bluntly: backups are necessary because “human and machine processing errors are very likely to generate corruption as time passes, and also the physical media where the data is stored is inevitably destined to fail.” Your laptop’s SSD is not a backup of anything. It is one copy, on one device, that fails eventually and takes everything with it.

Two failure modes drive every recommendation in this guide:

- **Hardware and corruption.** SSDs and disks fail. So do partitions, filesystems, and the software that writes to them. A single disk with a single copy has a single point of failure.
- **Deliberate attack.** Ransomware and “ransomware-style” attacks specifically target backup data to eliminate recovery options. A backup that an attacker with your credentials can rewrite is not a backup.

The second mode is why this document is longer than “copy your files to a USB stick.” A backup strategy is only as good as its least protected copy and its last successful restore test.

2. The five backup layers, compared

Linux backup tools fall into five families. They solve different problems, and the single most common beginner error is treating one of them as the whole answer. The table below compares the families; the notes underneath give the detail.

Layer	Representative tools	What it protects	Verdict for a laptop
Whole-system image	Rescuezilla, Clonezilla, Timeshift	The operating system itself: partitions, bootloader, installed packages. Rescuezilla restores “the exact same state they were in when the last snapshot was taken.”	**Yes, but only for the OS.** Timeshift’s own maintainers state it “is NOT a backup tool and is not meant to protect user data” — home directories are excluded by default.
File-level snapshot / dedup	Restic, Borg, BorgBase, Kopia	Your personal data: documents, photos, code, keys, dotfiles. Deduplicated, encrypted, compressed snapshots, portable across local disk, SFTP and cloud backends.	**The core layer.** This is what you point at ~. Borg adds no storage-layer redundancy of its own, so keep the target on a real second device.
Set-and-forget GUI	Déjà Dup (GNOME), Kopia GUI	The same job as the layer above, with the policy pre-set. Déjà Dup “hides the complexity of backing up the Right Way (encrypted, off-site, and regular) and uses Restic behind the scenes.”	**Good default for a beginner.** Less good if you want fine control over retention windows, exclusion rules, or target types.
Sync / mirror	rsync, Syncthing, Nextcloud, cloud drives	Convenience: keeping directories the same in several places. Tools “simply keep directories synchronized between multiple locations/machines, in a ‘mirror’ fashion.”	**Never the only copy.** Mirror semantics propagate deletions: rsync’s `--delete` removes the destination copy too, and most tools do not keep old revisions.
Versioned archive host	BorgBase and similar hosted Borg services	A remote target that already lives offsite, with server-side storage outside your control. Encryption still depends on your client-side configuration.	**Useful offsite leg.** Combine with an append-only repository mode so a compromised client cannot delete history.

Notes on the layers

Whole-system images. Rescuezilla is a graphical, Clonezilla-compatible tool you boot from USB; it writes a disk image to an external drive as managed files. Clonezilla is the text-mode original, offering device-image and savedisk workflows with a Beginner mode. Timeshift is Linux Mint’s snapshot tool using rsync+hardlinks

or Btrfs, restorable from a live CD/USB — but it is for system files and settings only. A laptop user needs an image tool *and* a file-level tool; they are not substitutes.

Encryption of the target. Restic derives a repository key from your password; Borg offers encrypted-repository modes; Déjà Dup encrypts volumes with a symmetric cipher driven by a password created at first run; Kopia encrypts all repositories with a password you designate (AES256-GCM-HMAC-SHA256 by default) that “remains in your computer and is not sent anywhere else.” Practical rule: an unencrypted external drive in your bag is a liability, and an encrypted repo whose password lives only on the dead laptop is useless. Store that password in a password manager on a different device.

Immutability, for the technically inclined. Both restic and borg support append-only repositories. Borg enables it with `borg config <path> to <repo> append_only 1`, which means a hacked client “cannot delete backups on the server permanently.” Restic warns that if an attacker has full read, write and delete access to the repository, “the protection offered by append-only mode is naturally void” — so run forget/prune from a separate, well-secured client.

https://rescuezilla.com · https://clonezilla.org/show-live-doc-content.php?topic=clonezilla-live%2Fdoc%2F01_Save_disk_image

https://github.com/linuxmint/timeshift

https://restic.readthedocs.io/en/stable/040_backup.html · https://borgbackup.readthedocs.io/en/stable/quickstart.html

https://borgbackup.readthedocs.io/en/stable/faq.html · https://kopia.io/docs/faqs

https://apps.gnome.org/DejaDup · https://help.gnome.org/users/deja-dup/stable/index.html

https://wiki.archlinux.org/title/Synchronization_and_backup_programs

3. 3-2-1, 3-2-1-1-0, and why the last zero matters

The 3-2-1 rule

Three copies of your data, two different types of storage media, one copy stored offsite. The ArchWiki calls this “one common and generally effective method,” and CISA recommends it as a trusted guideline, stressing “a combination of on-site and remote backups” plus scheduled recovery tests.

The 3-2-1-1-0 variant

Two extra letters, and both of them exist to close holes in the plain rule:

- **The extra 1 — immutable or air-gapped copy.** One copy attackers cannot modify even with valid credentials, because ransomware “specifically target[s] backup data to eliminate recovery options.” On a laptop this is a disk you only connect during a restore, or an append-only repository.
- **The 0 — zero errors after verification.** Backups must be tested and confirmed restorable, because jobs “fail for many reasons — storage limits, misconfigurations, software bugs — and often go unnoticed in a standard 3-2-1 setup.”

Real verification means testing the actual recovered data, not merely that a machine booted: “A server can start while its underlying database is corrupted or partially encrypted by dormant malware.” On a laptop the practical zero is simple: open a snapshot and confirm your documents are really there. That is the whole test.

What each layer looks like in practice

Slot	Role	Implementation	Offsite?
Copy 1	Live working set	The laptop’s own disk	No — this is the thing being protected
Copy 2	Local snapshot	External USB SSD or HDD with a Restic repository	Yes
Copy 3	Offsite	A drive at another address, or a cloud/SFTP backend	Yes
Copy 4 (immutable)	Append-only or offline	A disk connected only during restores	Yes
Verification	The zero	Monthly restore drill and `restic check --read-data`	Required

https://wiki.archlinux.org/title/System_backup

https://www.cisa.gov/audiences/small-and-medium-businesses/secure-your-business/back-up-business-data

https://objectfirst.com/guides/data-backup/3-2-1-backup-rule-and-strategy · https://www.datto.com/blog/3-2-1-1-0-backup-rule

4. Eight mistakes beginners make

1 Backing up to the same physical disk

This defeats the purpose. The ArchWiki's list of methods keeps snapshots on the same filesystem by default, and even Timeshift warns that "for best results the snapshots should be saved to an external (non-system) partition." A disk that dies takes both copies.

2 Assuming the backup tool means you have a backup

The most common and most damaging error. Mirror/sync semantics propagate deletions and mutations: rsync's `--delete` "makes sure that, when you delete a file from the source directory, it is ALSO deleted from your destination directory," and Restic even documents `restore --delete`. Anything that encrypts your files in place — including ransomware — gets faithfully snapshotted as encrypted.

3 Never verifying or restoring

Restic explicitly says "Now is a good time to run `restic check` to verify that all data is properly stored in the repository. You should run this command regularly," and that `--read-data` additionally verifies actual pack-file contents on disk. The ArchWiki says it is "highly encouraged to automate backups and test the recovery process to ensure everything works as intended."

4 Silently excluding dotfiles and hidden directories

A large share of what makes your setup yours lives in hidden paths — `~/.config`, `~/.ssh`, `~/.gnupg`, `~/.local/share`. Déjà Dup users have historically asked specifically how to get `~/.gnupg` into the backup. Test your exclusion rules by listing snapshot contents, not by assuming.

5 Backing up caches and rebuildable artifacts

Borg's own recommended daily script excludes `home/*/.cache/*` and `var/tmp/*`. On Arch you can also just keep a package list — "Maintain a list of all installed packages so that if a complete re-installation is inevitable, it is easier to re-create the original environment" — which makes backing up the package cache pointless.

6 No restore plan, and no idea how long a full restore takes

The ArchWiki documents how to make a backup actually bootable, including updating the backup's `/etc/fstab` and regenerating the bootloader config. CISA advises testing both full and partial restores. Know in advance whether your restore path needs an internet connection, an install USB, and your backup password.

7 Ignoring backup exit codes

Restic returns 0 on full success, 1 on fatal error with no snapshot created, and 3 when some source files could not be read (an incomplete snapshot). A nightly job that only mails you on non-zero will quietly ignore exit code 3 — which is the code that means your backup is incomplete.

8 Running out of space mid-backup

Restic "currently assumes that your backup repository has sufficient space"; running out leaves the repository partly written but no new snapshot. Borg warns that if it truly runs out of space "it can be hard or impossible to free space, because Borg needs free space to operate — even to delete backup archives."

https://wiki.archlinux.org/title/System_maintenance#Backups · https://wiki.archlinux.org/title/System_backup

https://restic.readthedocs.io/en/stable/040_backup.html · https://restic.readthedocs.io/en/stable/045_working_with_repos.html · https://restic.readthedocs.io/en/stable/050_restore.html

https://superuser.com/questions/156664/what-are-the-differences-between-the-rsync-delete-options

https://fedoramagazine.org/easy-backups-with-deja-dup

5. The recommended ten-step workflow

Do these in order. The order matters: verification before scheduling, and the OS layer last, because it is the layer you can rebuild.

Step 1 Decide what matters

Write down the irreplaceable things: documents, photos, code, SSH and GPG keys, browser bookmarks, password manager database, `~/.config`. Ignore the OS; you can reinstall that.

Step 2 Get a target on a different physical device

An external USB SSD or HDD you plug in, plus one offsite copy (cloud, SFTP, or a drive at another address). Two media types, one offsite — that is 3-2-1.

Step 3 Install restic

Debian/Ubuntu: `sudo apt install restic`. Fedora: `sudo dnf install restic`. Arch: `sudo pacman -S restic`. Debian-family also offers `sudo apt install borgbackup` if you prefer Borg.

Step 4 Initialise the repository on the external drive

Find the mount path first with `lsblk -f` or `ls /media/$USER/`, then:

Step 5 Write one backup command with sane excludes

This covers your dotfiles (restic backs up hidden files unless you exclude them) while skipping the caches Borg also excludes:

Step 6 Verify it worked, immediately

Run `restic snapshots` to confirm a snapshot exists, then `restic check` for structure and `restic check --read-data` for actual content. Then do a real restore drill and open some of your actual documents. This is the “0” in 3-2-1-1-0.

Step 7 Schedule it

Restic has no built-in scheduler; it “does not have a built-in way of scheduling backups, as it’s a tool that runs when executed rather than a daemon,” and points at systemd timers, cron, or resticprofile. ArchWiki’s template puts units at `/etc/systemd/system/backup.timer` with `OnCalendar=weekly` and `Persistent=true`, and stresses that “Backups that are only manually created are rarely up to date when they are needed.” If your target is over the network, the template’s `RestrictAddressFamilies` must be extended to `AF_UNIX AF_INET AF_INET6`.

Step 8 Apply a weekly retention policy

Always try `--dry-run` first. Note that pruning locks the repository so backups cannot run concurrently, and that `restic check` is advisable afterwards.

Step 9 Add the OS-recovery layer

Install Timeshift for system rollback, and take a Rescuezilla image to a second drive before major changes such as a disk swap or reinstall. Remember Timeshift does not cover your home directory.

Step 10 Set a recurring restore test

Once a month, repeat Step 6, plus one full-boot check of the Rescuezilla image on spare hardware or a spare partition. Write down how long it takes — that number is your real recovery time objective.

Command reference for Steps 4–6 and 8

```
export RESTIC_REPOSITORY=/media/$USER/BACKUP/laptop-restic && restic init

restic backup ~ \ --exclude '*.cache' \ --exclude '*/.cache/*' \ --exclude
'*/.thumbnails*' \ --exclude '*/.local/share/Trash/*' \ --tag daily

restic snapshots  &&  restic check  &&  restic check --read-data

mkdir -p /tmp/restore-test && restic restore latest --target /tmp/restore-test

restic forget --keep-daily 7 --keep-weekly 4 --keep-monthly 12 --prune   (try --dry-run
first)
```

https://restic.readthedocs.io/en/stable/020_installation.html

https://restic.readthedocs.io/en/stable/040_backup.html

https://restic.readthedocs.io/en/stable/050_restore.html

https://restic.readthedocs.io/en/stable/060_forget.html · https://restic.readthedocs.io/en/latest/060_forget.html

https://wiki.archlinux.org/title/System_backup · https://borgbackup.readthedocs.io/en/stable/quickstart.html

https://github.com/linuxmint/timeshift · https://rescuezilla.com

6. Limitations and caveats

This guide was assembled from documentation and vendor explainer pages. Readers should know exactly where the evidence is strong and where it is not.

- **No primary standards document for 3-2-1-1-0.** CISA's own pages endorse plain 3-2-1 and never mention the "1-0" extension. The 3-2-1-1-0 framing comes from vendor explainer pages (Datto, Object First, NovaBACKUP) rather than a standards body; they agree with each other but are not neutral parties.
- **The CISA pages are business-oriented.** They were mapped onto a personal-laptop context rather than written for consumers.
- **Déjà Dup GUI steps are unconfirmed.** GNOME's help index resolves but the individual "Backing Up" and "File Recovery" pages returned 404, so exact current GUI steps could not be confirmed from GNOME directly. Older Déjà Dup blog posts (Fedora Magazine, Duplicity era) are stale on the backend; GNOME now uses Restic.
- **No package or version verification.** Distro package names and versions for Timeshift, Rescuezilla, Clonezilla, Borg and Kopia were not verified.
- **Nothing was executed or benchmarked.** No command in this document was run. Restore-time figures (Rescuezilla's "within a few minutes") are vendor marketing copy, not benchmark data.

7. Sources

All URLs were taken from the research brief underpinning this document.

ArchWiki

- https://wiki.archlinux.org/title/System_maintenance#Backups
- https://wiki.archlinux.org/title/System_backup
- https://wiki.archlinux.org/title/Synchronization_and_backup_programs

Restic documentation

- https://restic.readthedocs.io/en/stable/020_installation.html
- https://restic.readthedocs.io/en/stable/040_backup.html
- https://restic.readthedocs.io/en/stable/045_working_with_repos.html
- https://restic.readthedocs.io/en/stable/050_restore.html
- https://restic.readthedocs.io/en/stable/060_forget.html
- https://restic.readthedocs.io/en/latest/060_forget.html

Borg documentation

- https://borgbackup.readthedocs.io/en/stable/quickstart.html
- https://borgbackup.readthedocs.io/en/stable/faq.html
- https://borgbackup.readthedocs.io/en/stable/usage/notes.html

Other tools

- https://rescuezilla.com
- https://clonezilla.org/show-live-doc-content.php?topic=clonezilla-live%2Fdoc%2F01_Save_disk_image
- https://github.com/linuxmint/timeshift
- https://kopia.io/docs/faqs
- https://kopia.io/docs/getting-started
- https://apps.gnome.org/DejaDup

- https://help.gnome.org/users/deja-dup/stable/index.html
- https://fedoramagazine.org/easy-backups-with-deja-dup

Guidelines and vendor explainers

- https://www.cisa.gov/audiences/small-and-medium-businesses/secure-your-business/back-up-business-data
- https://www.cisa.gov/audiences/state-local-tribal-and-territorial-government/secure-us-sltt/back-government-data
- https://objectfirst.com/guides/data-backup/3-2-1-backup-rule-and-strategy
- https://www.datto.com/blog/3-2-1-1-0-backup-rule

Other

- https://superuser.com/questions/156664/what-are-the-differences-between-the-rsync-delete-options